

Listening to the City

What does Melbourne talk about online, how does it feel about it, and who holds the conversation together?

A social-media neighbourhood watch built from the r/melbourne subreddit, combining topic modelling, sentiment analysis and reply-network analysis.

Research and report produced for: Assignment 2, COSC 2671 (Social Media and Network Analysis)

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Group: Postgraduate Group 33

Code repository: <https://github.com/lokesh-suroshe/smna-melbourne-group33>

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Introduction

Everyone who lives in a city has a rough sense of what people around them complain about and celebrate, but that sense is hard to pin down. Council surveys are slow and reach only a small group, the census is years apart, and the news tends to cover events rather than the steady hum of everyday life. We wanted to see whether a city's social media could fill that gap - whether, by reading what residents post, we could build something like an informal neighbourhood watch that shows what Melbourne is talking about, how it feels about each subject, and who keeps those conversations going.

We chose the r/melbourne subreddit as our window into the city. It is one of the largest Australian city subreddits, it is almost entirely text, and - importantly for us - every comment records what it is replying to, which gives us a ready-made social network on top of all that text. That combination is what made it suitable: we could do proper natural language processing (NLP) on the posts and proper network analysis on the replies, using the same dataset for both.

We broke our investigation into two parts, mirroring the two halves of the course. Part A - the network - looks at who talks to whom, what communities form, and who is influential. Part B - the content - looks at what is actually being said: the topics, the sentiment, and where in Melbourne the discussion points. We then bring the two together to ask whether the social structure and the subject-matter line up. Our guiding questions were: (1) what are the dominant topics and how do they change over time; (2) how does sentiment vary by topic and by suburb; and (3) who is central or acts as a bridge in the reply network, and do the network communities match the topic communities?

Before starting we agreed on what would make our results convincing rather than just a pile of charts: a coherent set of recognisable topics with their shares quantified; a sentiment breakdown by topic and suburb with enough volume to mean something; a directed weighted network analysed with several centrality measures and community detection, interpreted rather than just ranked; a numeric test of whether communities align with topics; and at least one practical insight a council or newsroom could actually use.

Background and Data

The idea of mining a city subreddit for civic signal is not ours alone - researchers and councils have long used social media to track public mood and emerging issues. What we wanted to add was the link between the text and the social structure: not just what is said, but whether the people saying similar things actually form connected communities, and who sits at the centre of them.

Our data is the public **r/melbourne corpus** distributed through the Cornell **ConvoKit** toolkit. We chose a curated public corpus rather than scraping the live site so that our work is fully

reproducible and involves no private data or API credentials. ConvoKit packages Reddit threads as conversations of utterances, each carrying the author, the parent it replies to, the text and a timestamp - exactly the fields we needed.

After flattening the corpus we worked with **2,378** root posts, **57,622** comments and **12,772** distinct users, covering **24 November 2009 to 31 October 2018**. We are upfront that this is a historical window: our findings describe r/melbourne over that period, not the city today.

We used the author field for network identity, the reply-to field for edges, the post and comment text for all the NLP, the timestamp for the over-time analysis, and the conversation id to keep comments grouped under their thread. The two main limitations we kept in mind throughout are that Reddit users are not a representative slice of Melbourne (they skew younger and more online), and that to keep the interaction network meaningful we focused on the actively-replying core rather than every account that ever posted once - we return to what that means in Part A.

Part A - The Network

Purpose

The aim of Part A was to understand the shape of the r/melbourne community: who replies to whom, whether distinct groups form, and who the influential or connecting users are. We expected, going in, that a city subreddit might split into topic-based camps - a transport crowd, a housing crowd, a politics crowd - each fairly separate. Part of the interest of our results is that this expectation turned out to be wrong in an informative way.

Building the network

We built a directed, weighted reply network. Each node is a user, and we draw an edge from user A to user B whenever A replies to B. We made it directed because a reply has a clear sender and receiver, and we care specifically about who is replied to, which is a sign of attention received. We weighted edges by how many times one user replied to another, so a repeated back-and-forth counts for more than a single passing reply. We removed bots and deleted accounts (AutoModerator, deleted and similar), and - importantly - we sampled whole conversations rather than individual comments so that every reply still points to a real parent in our data, keeping the network intact.

We then restricted the network to users with a total degree of at least two, to focus on the people actually holding conversations rather than the long tail of one-time posters. This left a core network of **373 users and 380 directed reply edges**, with a density of 0.0027. This is a deliberately conservative network: it is much smaller than the 12,772 total users because most of those users posted only once or twice and never entered a back-and-forth. We see this core as the engine room of the subreddit. A different threshold would change the absolute numbers, but the roles we describe below should be robust to that choice.

Who is influential? Three views of centrality

Rather than reporting a single top-users list, we computed three different centrality measures, because each captures a different kind of importance and the disagreements between them are where the interesting findings hide. Weighted in-degree measures raw attention - how many replies you receive. Betweenness measures brokerage - how often you sit on the path between other users, i.e. whether you bridge otherwise-separate conversations. PageRank measures influence that is itself weighted by the influence of the people replying to you. The top five users on each were:

#	In-degree (attention)	Betweenness (bridges)	PageRank (influence)
1	zozozozozoz	pressbutton	ratrl3
2	atregent	zozozozozoz	lemondrop__
3	sahrey	cameronm	zozozozozoz
4	ninjabk	Steve00	PM_ME_YOUR_CHESTHAMS
5	cayal3	continental-drift	illtakephotos

Figure 5 — Reply network (top 300 users by PageRank, colour = community) (RQ3)

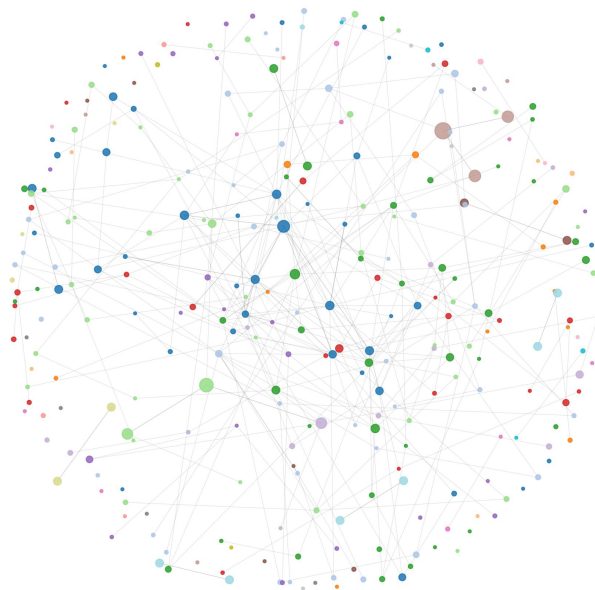


Figure A1 - The r/melbourne reply network (top users by PageRank). Node colour shows the community a user belongs to and node size is proportional to PageRank.

What we found is that the three lists only partly agree, and the pattern is telling. In-degree and betweenness overlap heavily - seven of the top ten are shared - which says that the users who attract the most replies are largely the same people who bridge different conversations. There is a recognisable core of visible, active participants who both pull in responses and stitch threads together. The user zozozozozoz is the clearest example, ranking at or near the top of all three measures.

PageRank tells a subtler story: it overlaps with in-degree on only four of ten. Users like `ratrl3` and `lemondrop__` top PageRank without being the most-replied-to overall, which means they receive replies from other influential users rather than simply a high volume of replies - a more quality-over-quantity kind of influence. Going the other way, `pressbutton` tops betweenness but is not the most influential on PageRank, which marks them as a connector or broker rather than a hub. The point we want to make here, and the reason we did not just report one ranking, is that attention, brokerage and influence are genuinely different roles in this community, and the same person rarely holds all three.

Communities - and a surprise

Running Louvain community detection on the network found **112 communities with a high modularity of 0.735**. A modularity that high means the network really does break cleanly into well-separated clusters - the largest has 37 members, with a long tail of smaller groups. So far this matched our expectation that `r/melbourne` would be a community of communities.

The surprise came when we checked whether those clusters were organised by topic. We gave each user their dominant topic (the subject they post about most) and compared that grouping against the Louvain communities using two standard agreement measures. Both came back low: **normalised mutual information of 0.30 and an adjusted Rand index of just 0.045**. In plain terms, the network clusters strongly, but it does not cluster by subject. The people who reply to each other are largely not the people who talk about the same things.

We think this is the most interesting result in Part A. We had expected topic-based echo chambers - a transport group, a housing group - and instead found that the same social clusters discuss a broad mix of subjects. The community structure is driven by who interacts with whom (shared threads, social ties, timing) rather than by shared interest. That is the signature of a general-purpose local community rather than a set of single-issue silos, and it is a genuinely different conclusion from the one we set out expecting.

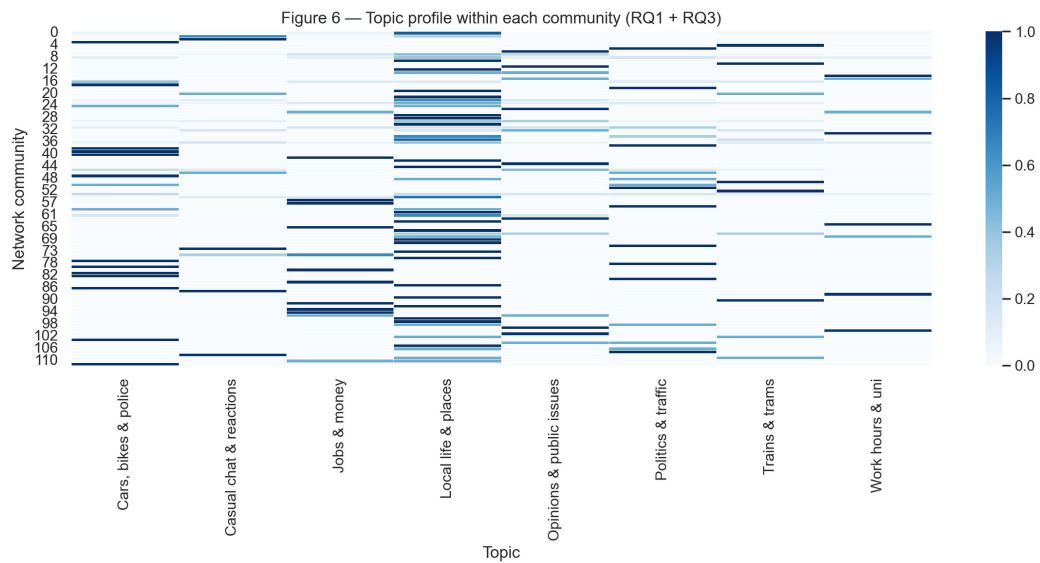


Figure A2 - Topic profile of each network community. If communities were topic-based we would expect one dark cell per row; instead most communities spread across several topics, and the popular Local life & places column lights up almost everywhere - the visual signature of the low NMI/ARI scores.

Part B - The Content

Purpose

Having mapped the social structure, Part B turns to what is actually being said. We used topic modelling to find the dominant subjects, sentiment analysis to gauge the mood of each, and a simple suburb-matching step to see whether sentiment varies across Melbourne geographically. We expected everyday life and local recommendations to dominate, and complaints (transport, traffic, policing) to be the most negative.

What Melbourne talks about

We modelled topics with Latent Dirichlet Allocation over a bag-of-words and bigram representation, settling on eight topics after checking that they read coherently. We labelled each topic by reading its top keywords and a sample of representative posts. In descending volume the eight topics are: Local life & places (9,444 documents), Opinions & public issues (6,505), Jobs & money (6,151), Cars, bikes & police (5,576), Casual chat & reactions (5,143), Trains & trams (5,116), Politics & traffic (4,706) and Work hours & uni (4,491).

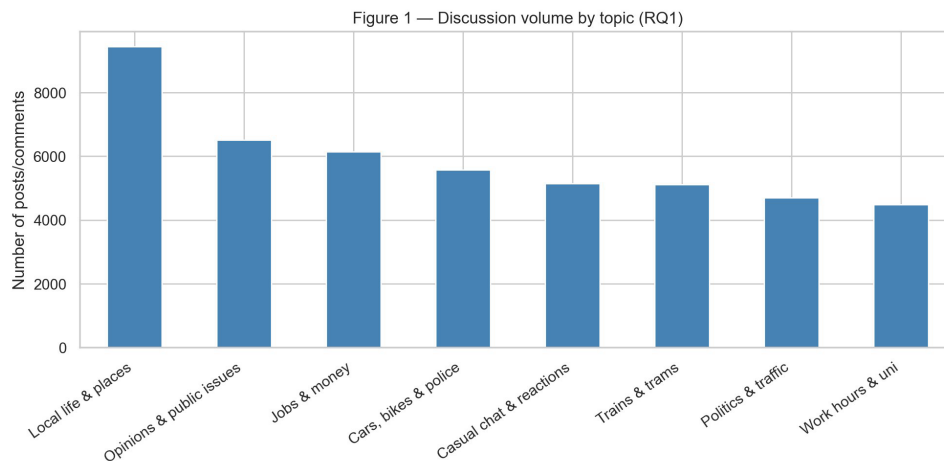


Figure B1 - Discussion volume by topic.

Everyday life clearly dominates: Local life & places is half as large again as the next topic, which fits the character of a city subreddit where recommendations, questions and local chat outweigh single-issue debate. Looking at how this changes over time, the early months are volatile simply because volume was low, but from around 2012 the mix settles down, with Local life & places consistently the largest share and the other topics spread fairly evenly. To us this says r/melbourne matured into a stable, general-purpose community rather than one dominated by any recurring issue - which, satisfyingly, agrees with the Part A finding that the network is not organised around topics.

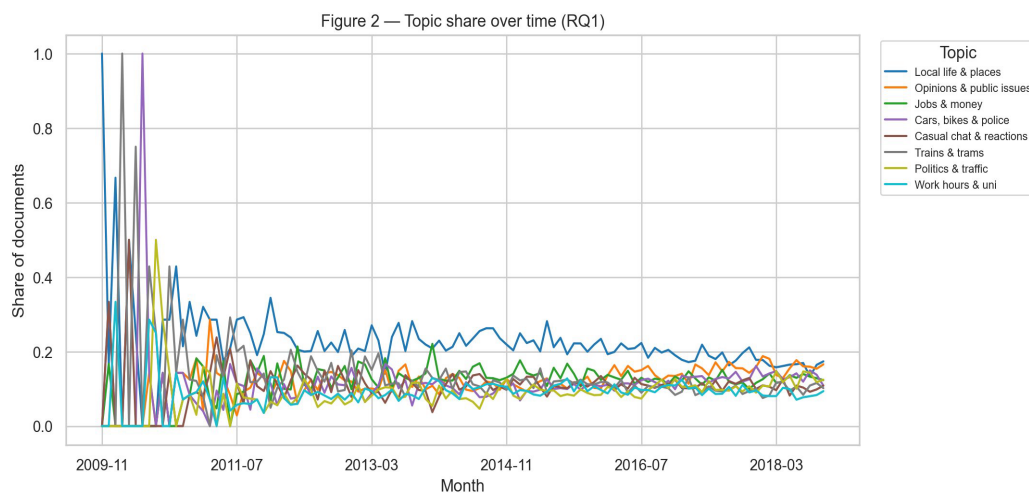


Figure B2 - Topic share over time.

How Melbourne feels

We scored every post and comment with VADER, a rule-based sentiment model built for social-media text, giving a polarity from -1 (negative) to +1 (positive), then averaged within each topic. The results match intuition almost too well. **Cars, bikes & police is the only net-negative topic** (mean -0.046) - unsurprising given it captures complaints about driving, cyclists and policing. At the positive end sit Local life & places (+0.285) and Jobs & money (+0.272), full of recommendations and good-news posts, while the civic-debate topics

(Opinions & public issues +0.048, Politics & traffic +0.045) sit close to neutral, consistent with mixed argument.

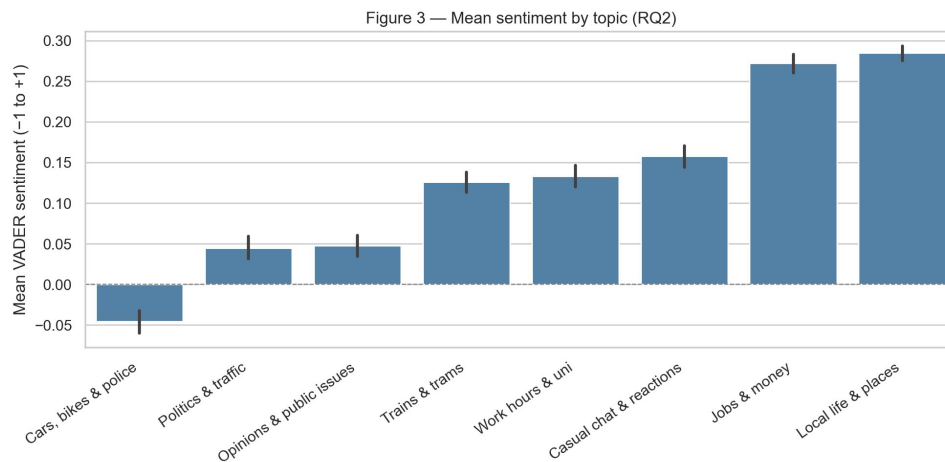


Figure B3 - Mean sentiment by topic (with confidence intervals).

Mapping sentiment to place, we matched posts against a list of Melbourne suburb names (removing the ambiguous word city and keeping only suburbs with at least ten mentions). Every well-mentioned suburb scores net-positive, but there is a clear gradient: inner and bayside suburbs such as Sunshine, Williamstown, Northcote and Collingwood are discussed most warmly, while outer-growth suburbs such as Pakenham, Frankston and Cranbourne sit lowest. We read this carefully - it almost certainly reflects how the subreddit's inner-city-leaning users talk about these places rather than any objective measure of the suburbs, and low-mention suburbs are less reliable. It is an indicative finding, not a verdict on anywhere.

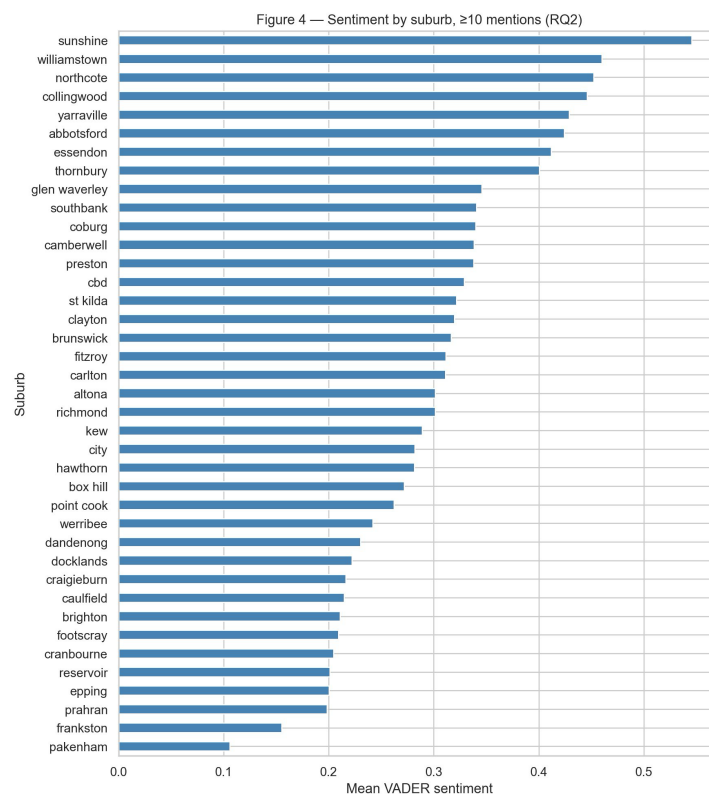


Figure B4 - Mean sentiment by suburb (suburbs with at least ten mentions).

Bringing It Together

Pulling Part A and Part B into one picture: r/melbourne's conversation is dominated by everyday local life, and the only consistently negative subject is cars, bikes and policing. Influence in the community is not held by a single set of top accounts but splits into distinct roles - high-attention hubs, bridge users, and PageRank-influential users trusted by other influential users. And although the network is strongly clustered, those clusters are organised socially rather than by topic, so the subreddit behaves as one broad community discussing many things rather than a collection of echo chambers.

The practical upshot - the insight we set ourselves the task of finding - is this. Because the most negative discussion concentrates in one identifiable topic (Cars, bikes & police) and the users who bridge conversations are a small, nameable set, a council or local newsroom wanting early warning of emerging road-safety or policing grievances would not need to monitor the whole subreddit. Watching that single topic and that handful of high-betweenness bridge users would give cheap, early signal, and the suburb gradient hints at where to look first. That is a concrete, low-cost monitoring strategy that falls directly out of combining the text and the network, which neither half could have produced alone.

Honest limitations

- The corpus runs 2009-2018, so everything here is historical; the topic mix may well have shifted since.
- r/melbourne is not Melbourne - the users are an online, English-speaking, inner-city-leaning subset, which especially colours the suburb-sentiment gradient.
- Our network is the degree-two-or-more core (373 users); a different threshold changes the absolute centrality values even if the relative roles hold.
- VADER misses sarcasm, which Reddit uses heavily, so individual sentiment scores are noisy even where the topic-level averages are stable.
- Suburb matching is approximate and low-mention suburbs are unreliable.

What we would do next

Given more time we would pull a recent live sample to see how much the topic mix has changed since 2018, swap VADER for a transformer sentiment model and try aspect-level sentiment to handle sarcasm, and test a few different network thresholds to confirm the centrality roles are stable. Adding a second source - local news comments, say - would let us check the suburb-sentiment pattern against something outside Reddit.

Conclusion

We set out to turn nine years of r/melbourne posts into a picture of what the community discusses, how it feels, and who holds the conversation together, and we met the criteria we set ourselves. We recovered eight interpretable topics with quantified shares, produced a sentiment breakdown by topic and suburb, built and analysed a directed weighted reply network with three centrality measures and community detection, tested community-topic alignment numerically, and drew a concrete monitoring insight from the combination. The result we did not expect - that the community is strongly clustered socially but mixed topically, so it is not a set of echo chambers - is the one we found most interesting, and it is exactly the kind of conclusion that only emerges when you analyse the text and the network together rather than separately.

Team Contributions and Repository

All code, the data sample, figures and team-management documents are in our repository: <https://github.com/lokesh-suroshe/smna-melbourne-group33>

Member	Main responsibilities	Approx. %
Lokesh Digamber Suroshe (s4160817, leader)	Data loading and preprocessing; network construction; centrality and community detection; Part A analysis and Figure A1.	50%
Yash Devidas Kadam (s4138198)	Topic modelling, sentiment and suburb extraction; Part B analysis and Figures B1-B4; the community-topic comparison write-up.	50%

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